

PH3244
Experiment - 5
The Cornu's method of determining the elastic constants of
a transparent material

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Synopsis

In this experiment, we use Cornu's method to determine the Young's modulus and Poisson's ratio of a given acrylic slab.

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I. THEORY AND PROCEDURE

A. Theory¹

The radius of curvature of plano convex lens R_0 is given by:

$$R_0 = \frac{d_n^2}{4n\lambda} \quad (1)$$

where the d_n^2 is the diameter of n^{th} fringe, λ is the wavelength used. The $d_n'^2$ and $d_n''^2$ along with d_n^2 can be seen properly in FIG - 1.

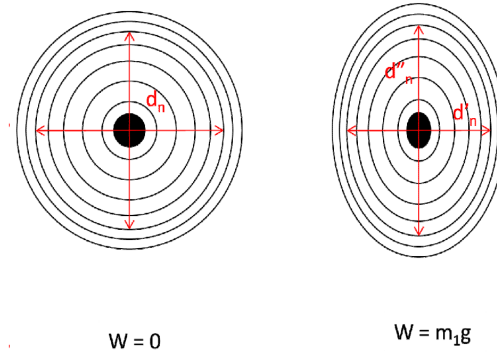


FIG. 1: Newton rings with and without bending shown, respectively, in the left and right panel.

The Young's modulus (Y) and the Possion's ratio (ζ) are calculated by:

$$Y = \frac{12m_1 g L R_1}{w h^3} \quad (2)$$

$$\zeta = \frac{R_1}{R_2} \quad (3)$$

Here m_1 is the mass on each side of the bar, g is the gravitational acceleration, L is the length of the beam from the pivot to a end, w and h are width and height of the rod respectively and can be seen from FIG - 2.

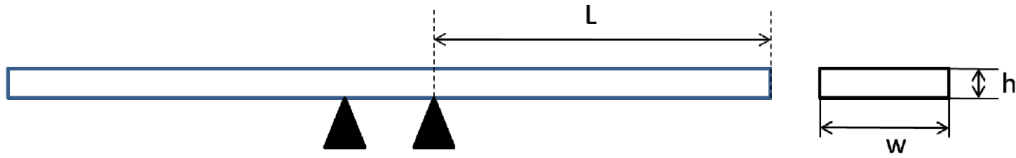


FIG. 2: Side views (the unloaded condition) of the beam whose elastic constants have to be determined. The beam is supported on two triangular pegs (black triangular shape objects in the figure)

¹ Please refer to [Cat (2026)] for a full examination of theory, only the important results will be discussed here.

Here R_1 and R_2 are obtained from the equations:

$$\frac{1}{R_1} = 4n\lambda \left(\frac{1}{d_n'^2} - \frac{1}{d_n^2} \right) \quad (4)$$

$$\frac{1}{R_2} = 4n\lambda \left(\frac{1}{d_n''^2} - \frac{1}{d_n^2} \right) \quad (5)$$

and are two different radius of curvature which can be seen from FIG - 3.

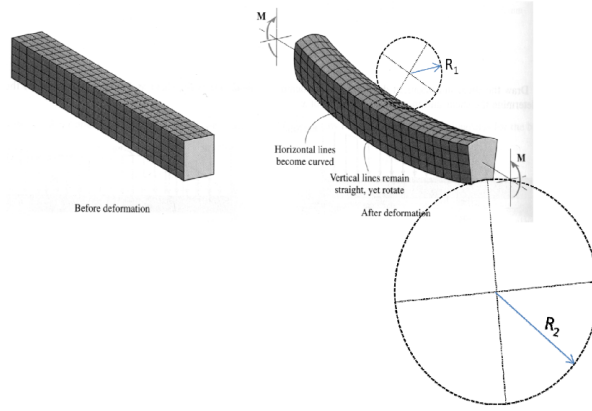


FIG. 3: Longitudinal bending (R_1) of a beam resulting in small lateral bending (R_2)

B. Procedure ²

From Equation - 1, 4 and 5, we can determine the values of R_0 , R_1 and R_2 from the slope obtained by plotting the graph of n vs different values from those radius of curvatures, from Equation - 2 and 3 we can determine what Young's modulus and Poisson's ratio of the given material is.

II. OBSERVATION

1. Slab parameters and other values³

$$\begin{aligned} w &= 41\text{mm} \\ l &= 125\text{mm} \\ h &= 3.7\text{mm} \\ g &= 9.8 * 10^3 \text{mm/s}^2 \\ \lambda &= 589 * 10^{-6} \text{mm} \end{aligned}$$

² Refer to the manual [Cat (2026)] for a look at procedure.

³ The least count is to be interpreted from the significant figures written

2. Fringe Readings⁴

Left		Right	
M.S.	C.S.	M.S.	C.S.
12.5	19	13.5	41
12.5	9	13.5	49
12.5	0	14.0	19
13.0	43	14.0	26
13.0	35	14.0	34
13.0	29	14.0	44
13.0	24	14.0	44

TABLE I: Readings for $m_1 = 0\text{g}$.

Left		Right		Top		Bottom	
M.S.	C.S.	M.S.	C.S.	M.S.	C.S.	M.S.	C.S.
13.5	18	15.0	31	11.0	6	11.5	14
13.5	7	14.0	40	10.5	37	11.5	25
13.5	0	14.0	49	10.5	29	11.5	33
13.0	43	14.5	6	10.5	21	11.5	39
13.0	38	14.5	12	10.5	15	11.5	46
13.0	32	14.5	17	10.5	9	12.0	4

TABLE II: Readings for $m_1 = 98\text{g}$.^a

^a On one side its 97g and the other 99g, hence taking average we get $m_1 = 98\text{g}$.

Left		Right		Top		Bottom	
M.S.	C.S.	M.S.	C.S.	M.S.	C.S.	M.S.	C.S.
13.5	13	14.0	21	10.0	40	11.0	6
13.5	3	14.0	31	10.0	28	11.0	18
13.0	44	14.0	39	10.0	20	11.0	25
13.0	37	14.0	46	10.0	13	11.0	33
13.0	32	14.5	2	10.0	7	11.0	40
13.0	27	14.5	7	10.0	1	11.0	46

TABLE III: Readings for $m_1 = 197\text{g}$.

Left		Right		Top		Bottom	
M.S.	C.S.	M.S.	C.S.	M.S.	C.S.	M.S.	C.S.
12.5	43	13.0	46	10.0	40	11.0	6
12.5	32	13.5	7	10.0	28	11.0	17
12.5	23	13.5	16	10.0	19	11.0	25
12.5	16	13.5	22	10.0	12	11.0	34
12.5	11	13.5	28	10.0	6	11.0	40
12.5	6	13.5	33	9.5	44	11.0	46

TABLE IV: Readings for $m_1 = 296\text{g}$.

Data taken on 16 Feb 2026.

III. CALCULATIONS, ANALYSIS AND CHARACTERISTICS

A. Sources of Error

1. Systematic Errors

1. The weight of the slab on itself.
2. Measuring chord instead of Diameter.
3. Unequal Weight on both sides.
4. Index of fringe being different than what is measured.

2. Random Errors

1. Human errors in measurement.
2. Uncertainty in instruments.
3. Shaking of table resulting in moving fringes.

B. Error Propagation

1. Calculating error from least count

To calculate the standard deviation from values we have measured with some least count, and the values we have applied digitally, we use the formula [Tewari (2016)]:

⁴ M.S. Stands for Main Scale Readings and C.S. Stands for Circulat Scale Readings

$$\sigma = \frac{\text{Least Count}}{\sqrt{12}} \quad (6)$$

and to calculate the standard deviation for values we have applied on analog instrument with some least count, we use the formula [Tewari (2016)]:

$$\sigma = \frac{\text{Least Count}}{\sqrt{24}} \quad (7)$$

2. Error propogation for R_0, Y and ζ

Now to calculate R_0 , from general error propogation formulas we see [Taylor (1997)] that:

$$\boxed{\sigma_{d_n^2} = \sqrt{2}d_n\sigma_{d_n} \implies \sigma_{d_n^2} = \frac{\text{Least Count}}{\sqrt{6}}} \quad (8)$$

Now for Y and ζ from Equation 2, 3 we see [Taylor (1997)] that:

$$\sigma_Y = Y \sqrt{\left(\frac{\sigma_{m_1}}{m_1}\right)^2 + \left(\frac{\sigma_g}{g}\right)^2 + \left(\frac{\sigma_L}{L}\right)^2 + \left(\frac{\sigma_{R_1}}{R_1}\right)^2 + \left(\frac{\sigma_w}{w}\right)^2 + \left(3\frac{\sigma_h}{h}\right)^2} \quad (9)$$

$$\sigma_\zeta = \zeta \sqrt{\left(\frac{\sigma_{R_1}}{R_1}\right)^2 + \left(\frac{\sigma_{R_2}}{R_2}\right)^2} \quad (10)$$

3. Error propogation for R_1

To calculate R_1 and R_2 we see that we need to plot n vs $\frac{x^2y^2}{|x^2-y^2|}$ (From Equation 4, 5). Let the uncertainty in d_n , d'_n , and d''_n be : σ_{d_n} , $\sigma_{d'_n}$, and $\sigma_{d''_n}$ respectively. For a function $f(x, y)$ with independent variables, the propagated error is

$$\sigma_f^2 = \left(\frac{\partial f}{\partial x}\right)^2 \sigma_x^2 + \left(\frac{\partial f}{\partial y}\right)^2 \sigma_y^2$$

Let

$$x = d_n, \quad y = d'_n$$

$$f = \frac{x^2y^2}{x^2 - y^2}$$

Using the quotient rule,

$$\frac{\partial f}{\partial x} = \frac{2xy^2(x^2 - y^2) - x^2y^2(2x)}{(x^2 - y^2)^2}$$

Simplifying,

$$\frac{\partial f}{\partial x} = \frac{-2xy^4}{(x^2 - y^2)^2}$$

Similarly,

$$\frac{\partial f}{\partial y} = \frac{2yx^4}{(x^2 - y^2)^2}$$

Substituting into the propagation formula,

$$\sigma_f^2 = \left(\frac{-2xy^4}{(x^2 - y^2)^2} \right)^2 \sigma_x^2 + \left(\frac{2yx^4}{(x^2 - y^2)^2} \right)^2 \sigma_y^2$$

$$\sigma_f^2 = \frac{4x^2y^8}{(x^2 - y^2)^4} \sigma_x^2 + \frac{4y^2x^8}{(x^2 - y^2)^4} \sigma_y^2$$

Thus

$$\sigma_f = \frac{2x^2y^2}{(x^2 - y^2)^2} \sqrt{y^4\sigma_x^2 + x^4\sigma_y^2}$$

Replacing x and y ,

$$\sigma_f = \frac{2d_n^2 d_n'^2}{(d_n^2 - d_n'^2)^2} \sqrt{d_n'^4 \sigma_{d_n}^2 + d_n^4 \sigma_{d_n'}^2} \quad (11)$$

4. Error propogation for R_2

Doing the exact same procedure as before and for $f = \frac{d_n^2 d_n'^2}{d_n'^2 - d_n^2}$ we get:

$$\sigma_f = \frac{2d_n^2 d_n'^2}{(d_n'^2 - d_n^2)^2} \sqrt{d_n'^4 \sigma_{d_n}^2 + d_n^4 \sigma_{d_n'}^2} \quad (12)$$

C. Plots and Calculation⁵

First step is to see weather the data makes iniuive sence or not. for this we plot Table 1 - 4 (we will not include uncertainty in this plot as we only want to see the characteristic) as follows:

⁵ Refer to [Cat (2026)] for the code used to plot and determine the slope.

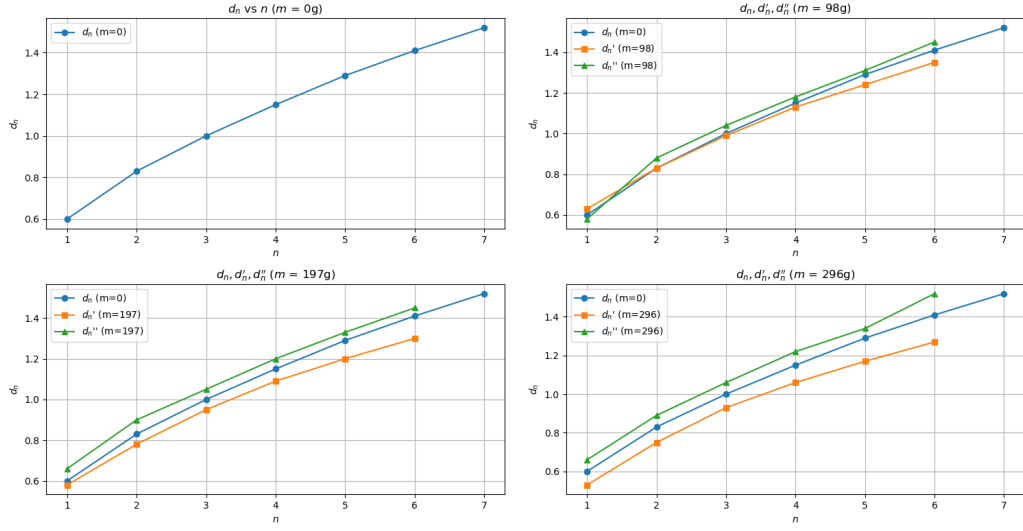


FIG. 4: Plot of Table 1 - 4 with d_n to compare

From this we see that other than Table - 2 (Top right figure) every other plots seems fine. In the Table two, thhr Hence in our later calcaultiosn we will not consider data from Table - 2 (The general expected order should be $d''_n > d_n > d'_n r$ but we dont see it for the initial values). Now using Table - 1 and Equation - 8 for uncertainty, we plot n vs d_n^2 as:

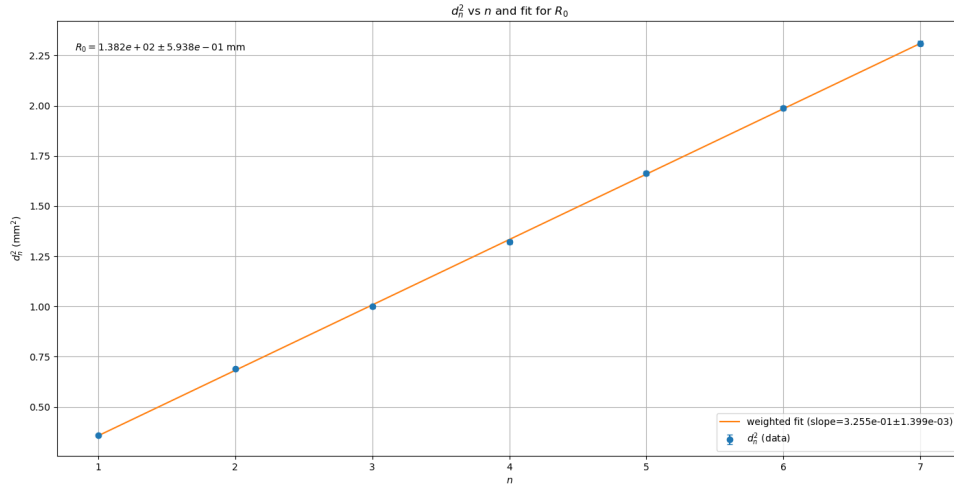


FIG. 5: Linear best fit with error to calculate R_0

Now to calcaulte the R_1 and R_2 we use the Equation 4, 5 to calculate it and Equation 10, 11 for its error. Doing it we get:

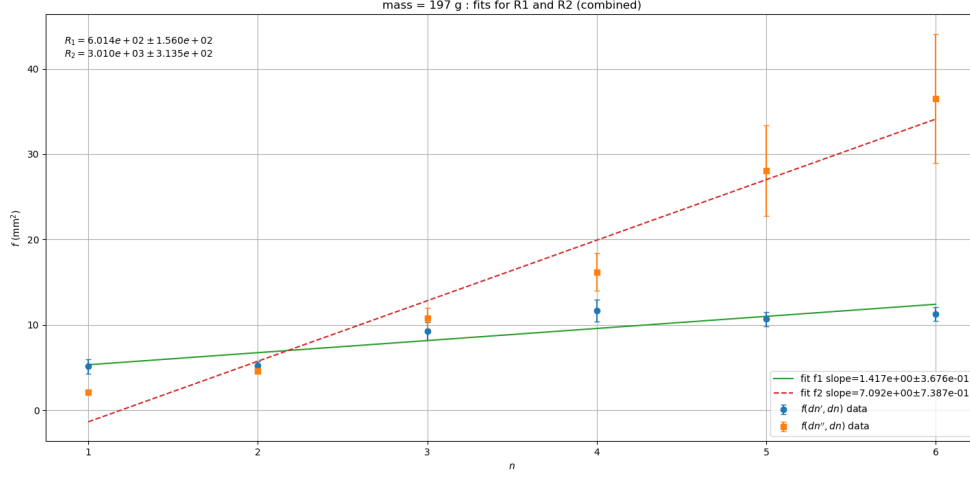


FIG. 6: Plot for $m_1 = 197\text{g}$ to calculate R_1 and R_2

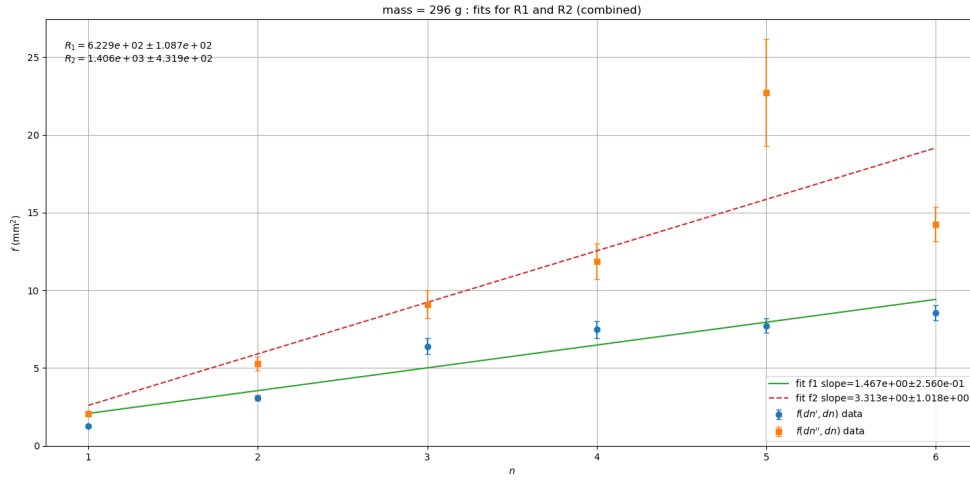


FIG. 7: Plot for $m_1 = 296\text{g}$ to calculate R_1 and R_2

From FIG. 6 and 7, we see that the calculated value of Y and ζ for each mass. Doing a weighted sum we get, $Y = 1.05145 \times 10^9 \pm 3.23540 \times 10^8 \text{ g s}^{-1} \text{ mm}^{-2}$.

IV. RESULT AND CONCLUSION

Hence after converting to SI, the Young's modulus will have unit of Pascals and Possion's ratio will be unitless. The final values we get are:

Young's modulus (Y) = $1.0 \pm 0.3 \text{ GPa}$ Poisson ratio (μ) = 0.325 ± 0.17

The error in these values along with the theoretical value will be discussed in Appendix A.

REFERENCES

- L. M. Cat, Ph3244 3rd year sem 6 physics lab repo: <https://github.com/laughingcatmeme/ph3244> (2026).
G. M. Tewari, Measurement uncertainty, International Workshop and Training Program on Good Food Laboratory Practices (2016).
J. R. Taylor, *An Introduction to Error Analysis* (1997).

Appendix A: Theoretical values and Uncertainty

Perspex (PMMA) acrylic typically exhibits a Young's modulus of 2.4–3.3 GPa (2400–3300 MPa) and a Poisson's ratio of 0.35–0.4. The cause of the deviation for Young's modulus is attributed to very small sample of masses taken. The Poisson's ratio fits well within the theoretical value although it has very high uncertainty again because of low sample size.

Appendix B: Improvements in procedure

The uncertainty can be decreased by taking more weight measurements. Furthermore to reduce human error we can take multiple measurements for each mass. We can also reduce some systematic errors, such as the weight of the slab, by measuring it and taking it into account for calculations. To have correct fringe index, we can take a picture to determine the peaks, this will be more accurate than eye and can also be double checked later.